

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

4 Week: 6 March 2017 / 6 Maart 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P4.38 (a) 6.00 rev/s (b) $1.52 \times 10^3 \text{ m/s}^2$ (c) $1.28 \times 10^3 \text{ m/s}^2$

P4.39 $v = 7.58 \times 10^3 \text{ m/s}$ and $T = 5.80 \times 10^3 \text{ s}$

P5.19 (a) $a = 5.00 \text{ m/s}^2$ at $\theta = 36.9^\circ$ (b) $a = 6.08 \text{ m/s}^2$ at 25.3°

P5.22 (a) 181° counter-clockwise from the positive x -axis (b) 11.2 kg

(c) 37.5 m/s (d) $\vec{v}_f = (-37.5\hat{i} - 0.893\hat{j}) \text{ m/s}$

P5.31 (a) FBD (b) 613 N

P5.34 Proof

P5.36 (a) $T_1 = 31.5 \text{ N}$, $T_2 = 37.5 \text{ N}$ and $T_3 = 49.0 \text{ N}$ (b) $T_1 = 113 \text{ N}$, $T_2 = 56.6 \text{ N}$ and $T_3 = 98.0 \text{ N}$

P5.37 $\vec{F}_1 = 8.66 \text{ N east}$

P5.47 3.73 m

P5.49 (a) 2.20 m/s^2 (b) 27.4 N

P5.60 (a) FBD (b) $\theta = 55.2^\circ$ (c) 167 N

P5.64 (a) FBD's (b) 2.31 m/s^2 , down for m_1 , left for m_2 , and up for m_3

(c) $T_{\text{left}} = 30 \text{ N}$, $T_{\text{right}} = 24.2 \text{ N}$ (d) Explanation